Lehmer's number sieve

West bay 4: wheels that sieve for factors · 14 November 2026

Babbage's Hall, west bay 4. A number sieve after Derrick Lehmer's (1926 and 1932): wheels of different numbers of places, all moved on together, with holes where a number passes a test. Where every wheel has an open hole at the reading place, a probe falls right through, and the machine stops: that number passes every test at once.

Lehmer's first sieve was made of bicycle chains, each loop as many links long as its test's modulus, with a rod standing out of the links for the remainders that pass. The rods closed small switches as the chains went round, and when every switch was closed at once, a signal ran the length of the machine. His 1932 sieve had gears of different sizes with a hole for each remainder, the holes not wanted plugged (with toothpicks), a bright light on one side and a photocell on the other: when the holes lined up, light reached the cell, and its signal, amplified, stopped the motor. This one is mechanical throughout: holes in printed wheels, pegs in the holes that fail (Lehmer's toothpicks), and a probe that falls by its own weight.

What it does

It factors a number N by Fermat's way. If x * x - N is a square, y * y, then N = (x - y)(x + y). Fermat tried x = the first whole number past the square root of N, then the next, and so on. The sieve does the trying. A square leaves only certain remainders when divided by m (by 8: 0, 1 or 4), so x * x - N can be a square only when x mod m is one of a few. On the wheel of m places, those holes are open and the rest are pegged. After x turns from 0, the hole under the probe on the wheel of m places is the one marked x mod m, so the probe falls right through only at an x that passes the test for all four wheels at once: 9, 8, 7 and 5 places, which between them repeat only every 2520 turns.

The sieve only says "maybe": each x it stops at must still be tried by hand (is x * x - N a square?). For N = 1241, starting at x = 36, it stops at 39 (39 * 39 - 1241 = 280, not a square), and at 45 (45 * 45 - 1241 = 784 = 28 * 28), so 1241 = 17 x 73. Over every 2520 turns, only 144 x get past all four wheels for 1241 (one in 17); the tour's first ten happen to hold two of them. With more wheels it would reject still more. Lehmer's chain sieve tried about 60 numbers a second, his gear sieve about 5000.

The machine

- **Four wheels** on one fixed axle, one above the other: 9 places at the bottom, then 8, 7 and 5. Each is a **Geneva star**: a wheel with a slot for each place. Beside them is the **drive shaft**, with the handle on top. Under each wheel the shaft carries a crank with a pin. Once a turn, the pin enters a slot, turns the star on by exactly one place, and leaves it. So one turn of the handle moves every wheel on by one place, whatever its size. Beside each star, the shaft carries a **locking disc** that rides in the star's curved edge between turns, so the star can't be turned out of step.
- **A detent** for each wheel, a printed leaf spring with a round nose that rests in the mouth of the slot opposite the drive, against both of its corners. It puts each star exactly on its place between turns. (The locking disc keeps a star within about a degree of its place: the detent puts it there. See `FINDINGS.md`.)
- **A ring of holes** in each wheel, one per place, marked with its number. **Pegs** close the holes of the numbers that fail.

- **The probe**, a steel rod at the reading place, above the stack, in a guide. An arm from it reaches over to the **cam** on top of the drive shaft, which lifts it before the wheels move and lets it down after they have stopped. It falls through as many wheels as have an open hole under it. Right through, it comes to rest in the **bell** at the bottom.

Using it

Choose a number with `N`, then run its tour with `T`, or turn the handle yourself with `G` or `Space`. The hand turns the handle a whole turn at a time. Watch the probe: it goes down into the stack as far as the first pegged hole. When it goes right through, `G` stops; the tour stops only at the true answer, carrying on past a false stop.

Where to look.

- **The board** on the table to the left shows what an operator would write up. N and x; under "mod 5", "mod 7", "mod 8" and "mod 9", the hole now under the probe on each wheel, green if it is open and red if it is pegged; how many wheels the probe went through (blank while the wheels move). When the probe goes right through, it shows x * x - N and whether that is a square, and if it is, y and the two factors. The board is the exhibit's, not part of the machine: nothing in the machine drives it.
- **The depth scale,** a post on the probe's guide beside its top, is the machine's own (printed). The pointer on the probe's top reads 0 to 4, the number of wheels the probe went through; 4 is right through, onto the bell.
- **The console** prints each turn's reading, and at each stop the working: x * x - N, and whether it is a square.

Key	
`Space`	one turn of the handle
`G`	go: turn until the probe falls right through (`G` again: stop)
`N`	choose the next number to factor: 1241, 1679, 1157, 2813 (re-pegs the wheels and sets them to its first x)
`R`	set the wheels back to the first x
`T`	the tour for the number chosen: it turns until the factors are found (it starts with 1241's)
`Z`	slow motion (1/4 speed) on and off

Rest the mouse on a wheel to see which hole is under the probe and which are open; on the probe or the scale, how far it has gone through; on the board, what it shows.

Sliders: `friction` (0.1 to 0.4 checked), `turn rate` (how fast the hand turns the handle, 0.25 to 1 turn a second checked), `speed`.

Running it

Open `lehmer-sieve.lua` in bpp, or run it inside the hall (`HALL_START=lehmer`). It starts with the tour, which prints each turn's reading and `RESULT lehmer-sieve tour: PASS` when every reading was right and the factors are found. At every turn it checks that the probe went exactly as far as the pegs say it should, and that the wheels, read on their own, show the x the hand has counted to.

Environment, for testing: `LS_TOUR=0` (no tour), `LS_N` (the number to factor), `LS_GO=1` (turn until the probe falls through, without the tour), `LS_FRICTION`, `LS_RATE`, `LS_LOG=1` (every turn's reading), `LS_LOG=2` (also the handle, the wheels' places, the probe and the detents every half second; `LS_LOGDT` sets how often), `LS_HZ` (steps a second, 2400), `LS_ITER` (solver passes, 20), `LS_REST=0` (never rest), `LS_REST_LOG=1`.

Once the hand is done and nothing moves, the machine rests: it stops stepping until the next key, which saves the processor and changes nothing it computes.

How it is built

- **One source.** `parts/make-geometry.py` (needs shapely) works out every moving part's outline: the stars with their slots, locking arcs and holes, the locking discs (a disc, less wherever its star passes in a turn), the cranks, the cam's profile and the detents. It writes them to `parts/geometry.scad`, which `parts/lehmer-sieve.scad` builds the printed parts from, and to `sieve-geometry.lua`, the same outlines cut into convex pieces for the simulation. What is printed is what was simulated.
- **Bodies.** Each wheel is one body on a hinge about the axle; its pegs are part of it (a peg not wanted is moved into the wheel's hub, where it touches nothing). The drive is one body on a hinge (the shaft, cranks, pins, locking discs and cam). Each detent is a body on a hinge at its leaf's root, turned back by the leaf's own stiffness (3EI/L, PLA's modulus), with a stop. The probe slides in its guide (a slider).
- **The hand** is a motor on the drive's hinge, steered a whole turn at a time. The wheels move only as the pins push them, and the probe only as the cam and gravity move it.
- **Idealised:** the hand stops turning when the probe falls right through (Lehmer's photocell and relay did that); here a person watching would. The wheels are set to their first x by hand (lift each off its axle, turn it, put it back), as the simulation does by placing them.
- **The board** is the exhibit's display, not the machine's: an operator would write the same things on a slate.

Printing and building

`./make-stls.sh` (needs OpenSCAD) writes every part into `stl/`, each turned to lie on the bed. The largest is the 9-place wheel, 150 x 152 mm.

Part	Print	Size (mm)	Notes
`wheel-9.stl`, `-8`, `-7`, `-5`	1 each	149 × 152 x 5.5 (the 9)	the numbers stand 0.5 mm proud: paint them, or change filament there
`peg.stl`	20	7.6 × 7.6 × 5	a push fit in the 8 mm holes; paint them red
`crank-9.stl` … `-5`	1 each	42 to 63 × 15 x 5	a 6 mm steel dowel pressed into the end, standing 6 mm up
`lock-9.stl` … `-5`	1 each	42 to 89 × 47 to 89 × 6	the locking discs
`cam.stl`	1	90 × 90 × 65	15% infill is plenty
`handle.stl`	1	44 × 14 × 25	
`detent-9.stl` … `-5`	1 each	10 to 23 × 37 to 42 × 5	PLA, 100% infill; the leaf is 1.2 mm, three lines of a 0.4 mm nozzle; its root goes on 6 mm past the hinge, into the post's slot
`detent-post-9.stl` … `-5`	1 each	24 to 57 tall	the leaf's root presses into the slot in its block
`probe-arm.stl`	1	79 × 12 × 13	the ball is glued into the socket under its end
`probe-pointer.stl`	1	20 × 9 × 3	on the probe's top: it reads the depth scale
`guide.stl`	1	99 × 152 × 12	printed on its side; the depth scale's marks stand 0.5 mm proud (paint them)
`washer.stl` x3, `washer-low.stl` x1		18 across	between the wheels on the axle

Part	Print	Size (mm)	Notes
`bell.stl`, `bearing.stl`, `shaft-washer.stl`, `axle-nut.stl`, `foot.stl` x4			the bearing goes under the base; the shaft washer on it, under the lowest crank

Also needed

- **Plywood,** 12 mm, 270 x 190 mm, for the base: `stl/cut-base.svg` has its outline and holes. Glue the four feet under its corners.
- **8 mm steel rod,** 65 mm, the wheels' axle (pressed into the base).
- **6 mm steel rod,** 135 mm, the drive shaft, with a flat filed 0.8 mm deep along it (the drive's parts have D-shaped bores, so that the pins, the locking discs and the cam go on at their angles). It turns in the bearing under the base and in the base.
- **4 steel dowels,** 6 x 10 mm, the cranks' pins.
- **5 mm steel rod,** 85 mm, the probe.
- **An 8 mm steel ball,** the follower under the probe's arm.
- Glue; a little dry PTFE on the slots, the locking arcs and the cam.

Assembly. Press the axle and the detent posts into the base, the shaft through the bearing and the base. Put the shaft washer on the shaft, then slide the drive's parts down it in order: for each wheel from the bottom, its crank, then its locking disc; then the cam. Each goes on the flat, and they stand on one another's hubs. Glue the handle on the top. Put the low washer on the axle, then the 9-place wheel, a washer, the 8, a washer, the 7, a washer, the 5, and the nut. Turn the handle slowly once to see each pin go into a slot and out again. Fit the detents (press each leaf into its post's slot; the nose drops into the mouth of the slot opposite the drive). Set the guide in the base, the probe through its sleeve, the arm on the probe with its ball on the cam. Then peg the wheels for a number (the console lists the open holes), set each wheel by hand so the hole for x mod m is under the probe, and turn.

See `FINDINGS.md` for how the numbers were arrived at.